

# CSE 4621: Machine Learning

# Decision Trees

## Lecture 10

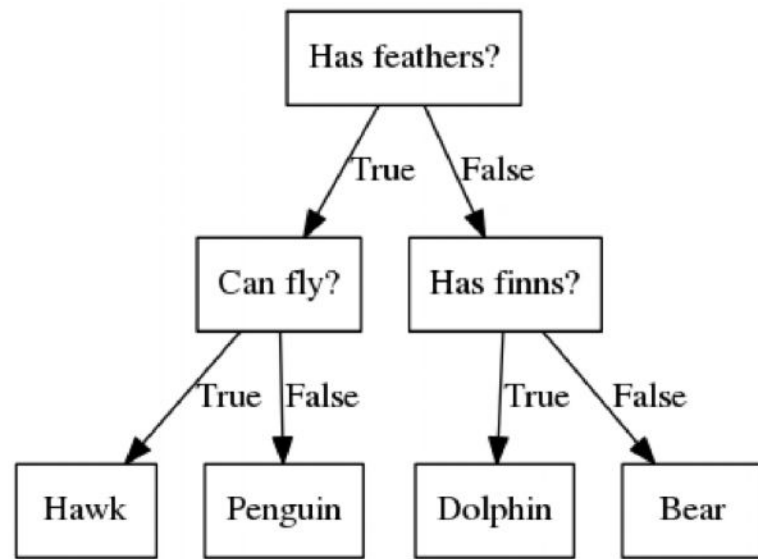
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# Lecture Outline

- Decision Tree
  - Classification Tree
  - Impurity measure
  - Entropy
  - Gini Impurity
  - Split and Sub-tree Generation
  - Information Gain
- Regression Trees

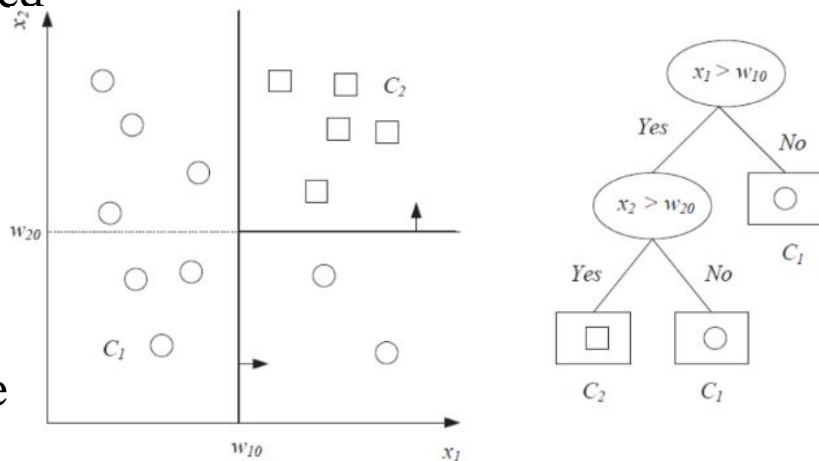
# Decision Tree

- A decision tree is a hierarchical data structure implementing the divide-and-conquer approach
  - It is a non-parametric method
  - Can be used for both classification and regression
- It is composed of internal decision nodes and terminal leaves
- A combination of multiple *if-else* statements



# Decision Tree

- Given an input, on each node, a test is applied and one of the branches is taken depending on the outcome
- Each decision node  $m$  implements a test function  $f_m(x)$  with discrete outcomes labeling the branches
- This is repeated recursively until a leaf node is hit



# Test Function

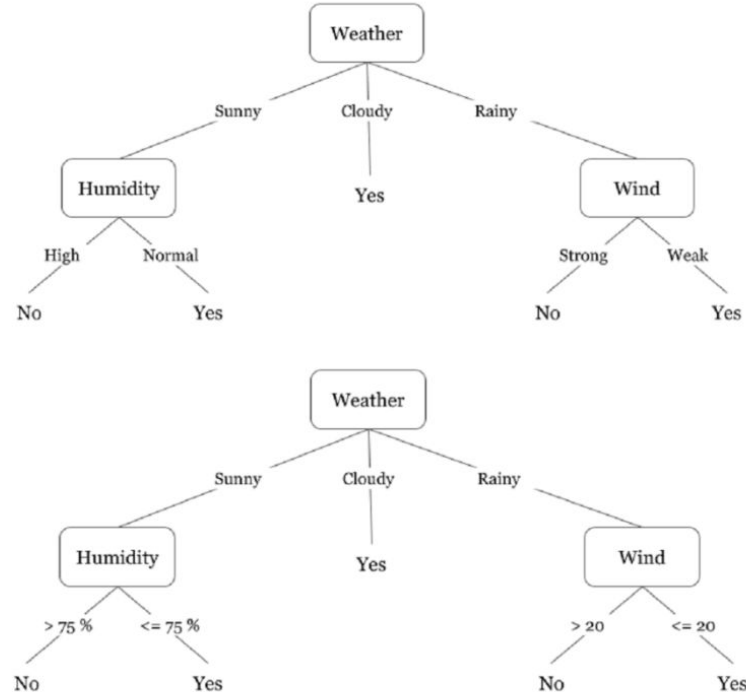
- Each test function  $f_m(x)$  defines a discriminant in the  $d$ -dimensional space, dividing it into smaller subsets
- Different tree families use different test models, producing different decision boundaries
- Hierarchical decisions can locate a region efficiently –  $\log(b)$  decisions in the best balanced case

# Univariate Tree

In a *univariate* tree, each internal node's test function uses only one of the input features (attributes)

## Example

| Day | Weather | Temperature | Humidity | Wind   | Play? |
|-----|---------|-------------|----------|--------|-------|
| 1   | Sunny   | Hot         | High     | Weak   | No    |
| 2   | Cloudy  | Hot         | High     | Weak   | Yes   |
| 3   | Sunny   | Mild        | Normal   | Strong | Yes   |
| 4   | Cloudy  | Mild        | High     | Strong | Yes   |
| 5   | Rainy   | Mild        | High     | Strong | No    |
| 6   | Rainy   | Cool        | Normal   | Strong | No    |
| 7   | Rainy   | Mild        | High     | Weak   | Yes   |
| 8   | Sunny   | Hot         | High     | Strong | No    |
| 9   | Cloudy  | Hot         | Normal   | Weak   | Yes   |
| 10  | Rainy   | Mild        | High     | Strong | No    |



# Impurity Measure

- In a classification tree, the goodness of a split is quantified by an impurity measure
- The function  $\phi(p, 1 - p)$  is an impurity measure, where  $p$  is the proportion of one class
- A split is pure if every example in each branch belongs to the same class
- A split is maximally impure if it contains a 50/50 mixture

Impurity rises as the classes become more balanced, then falls again

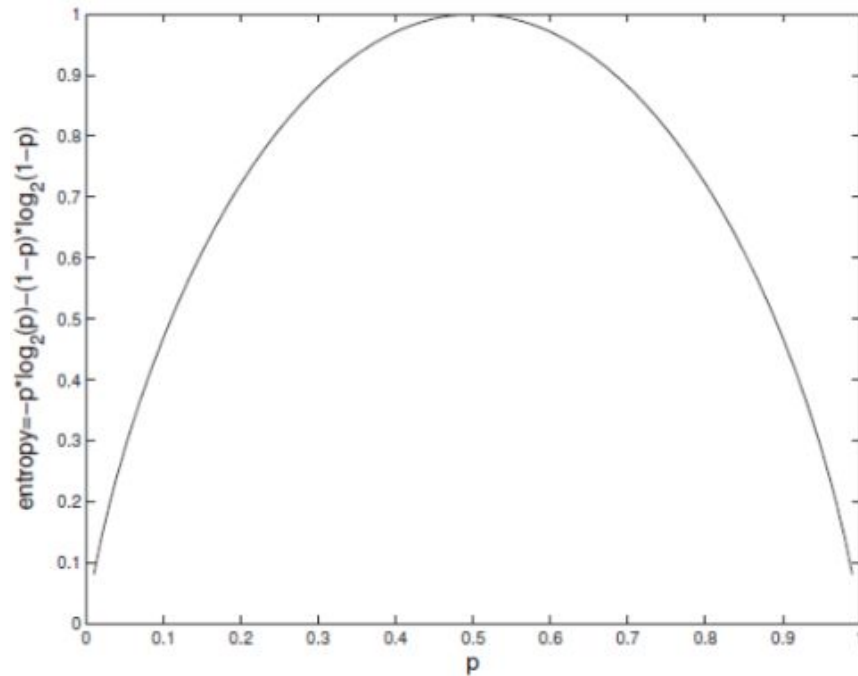
# Impurity Measure: Entropy

Our measure of impurity is entropy:

$$\phi(p, 1-p) = -p \log(p) - (1-p) \log(1-p)$$

For  $K$  classes:  $\mathcal{I}_m = - \sum_{i=1}^K p_m^i \log(p_m^i)$

- $m \rightarrow$  current tree node
- $K \rightarrow$  number of classes
- $p_m^i \rightarrow$  proportion of the samples in  $m$  that belong to class  $i$
- $I_m \rightarrow$  entropy of node  $m$





# Split and Sub-tree Generation

- If a node is not pure, evaluate candidate attributes and split positions
- Compute the impurity of the child nodes after each candidate split
- Choose the split with the minimum post-split impurity or the largest impurity decrease
- This is a greedy algorithm, locally choosing the most optimum split

The total impurity after a split:  $\mathcal{I}'_m = - \sum_{j=1}^n \frac{N_{mj}}{N_m} \sum_{i=1}^K p_m^i \log(p_m^i)$

Gain  $(S, T) = \text{Entropy}(S) - \text{Entropy}(S, T)$

# Example

- Sample # 14

| Play Golf(14) |    |
|---------------|----|
| Yes           | No |
| 9             | 5  |

Fig. Frequency Table

$$E(\text{PlayGolf}) = E(5,9)$$

$$= -\left(\frac{9}{14} \log_2 \frac{9}{14}\right) - \left(\frac{5}{14} \log_2 \frac{5}{14}\right)$$

$$= -(0.357 \log_2 0.357) - (0.643 \log_2 0.643)$$

$$= 0.94$$

| Outlook  | Temperature | Humidity | Windy | Play Golf |
|----------|-------------|----------|-------|-----------|
| Rainy    | Hot         | High     | FALSE | No        |
| Rainy    | Hot         | High     | TRUE  | No        |
| Overcast | Hot         | High     | FALSE | Yes       |
| Sunny    | Mild        | High     | FALSE | Yes       |
| Sunny    | Cool        | Normal   | FALSE | Yes       |
| Sunny    | Cool        | Normal   | TRUE  | No        |
| Overcast | Cool        | Normal   | TRUE  | Yes       |
| Rainy    | Mild        | High     | FALSE | No        |
| Rainy    | Cool        | Normal   | FALSE | Yes       |
| Sunny    | Mild        | Normal   | FALSE | Yes       |
| Rainy    | Mild        | Normal   | TRUE  | Yes       |
| Overcast | Mild        | High     | TRUE  | Yes       |
| Overcast | Hot         | Normal   | FALSE | Yes       |
| Sunny    | Mild        | High     | TRUE  | No        |

# Example

We need to calculate the entropy after each of the splits:

- $E(\text{PlayGolf}, \text{Outlook}) \rightarrow P(\text{Sunny})E(\text{Sunny}) + P(\text{Overcast})E(\text{Overcast}) + P(\text{Rainy})E(\text{Rainy})$
- $E(\text{PlayGolf}, \text{Temperature})$
- $E(\text{PlayGolf}, \text{Humidity})$
- $E(\text{PlayGolf}, \text{Windy})$

# Example

*$E(\text{PlayGolf}, \text{Outlook})$*

|         |          | PlayGolf(14) |    |   |
|---------|----------|--------------|----|---|
|         |          | Yes          | No |   |
| Outlook | Sunny    | 3            | 2  | 5 |
|         | Overcast | 4            | 0  | 4 |
|         | Rainy    | 2            | 3  | 5 |

Fig. Frequency Table

$$E(\text{Sunny}) = E(3,2)$$

$$\begin{aligned}
 &= -\left(\frac{3}{5} \log_2 \frac{3}{5}\right) - \left(\frac{2}{5} \log_2 \frac{2}{5}\right) \\
 &= -(0.60 \log_2 0.60) - (0.40 \log_2 0.40) \\
 &= -(0.60 * 0.737) - (0.40 * 0.529) \\
 &= \mathbf{0.971}
 \end{aligned}$$

$$E(\text{Overcast}) = E(4,0)$$

$$\begin{aligned}
 &= -\left(\frac{4}{4} \log_2 \frac{4}{4}\right) - \left(\frac{0}{4} \log_2 \frac{0}{4}\right) \\
 &= -(0) - (0) \\
 &= \mathbf{0}
 \end{aligned}$$

$$E(\text{Rainy}) = E(2,3)$$

$$\begin{aligned}
 &= -\left(\frac{2}{5} \log_2 \frac{2}{5}\right) - \left(\frac{3}{5} \log_2 \frac{3}{5}\right) \\
 &= -(0.40 \log_2 0.40) - (0.6 \log_2 0.60) \\
 &= \mathbf{0.971}
 \end{aligned}$$

$$E(\text{PlayGolf}, \text{Outlook}) = \frac{5}{14} E(3,2) + \frac{4}{14} E(4,0) + \frac{5}{14} E(2,3)$$

$$\begin{aligned}
 &= \frac{5}{14} 0.971 + \frac{4}{14} 0.0 + \frac{5}{14} 0.971 \\
 &= 0.357 * 0.971 + 0.0 + 0.357 * 0.971 \\
 &= \mathbf{0.693}
 \end{aligned}$$

# Example

- $E(\text{PlayGolf}, \text{Outlook}) = \mathbf{0.693}$
- $E(\text{PlayGolf}, \text{Temperature}) = \mathbf{0.911}$
- $E(\text{PlayGolf}, \text{Humidity}) = \mathbf{0.788}$
- $E(\text{PlayGolf}, \text{Windy}) = \mathbf{0.892}$
- $\text{Gain}(S, T) = \text{Entropy}(S) - \text{Entropy}(S, T)$
- $\text{Gain}(\text{PlayGolf}, \text{Outlook}) = \text{Entropy}(\text{PlayGolf}) - \text{Entropy}(\text{PlayGolf}, \text{Outlook})$   
 $= 0.94 - 0.693 = \mathbf{0.247}$
- $\text{Gain}(\text{PlayGolf}, \text{Temperature}) = \text{Entropy}(\text{PlayGolf}) - \text{Entropy}(\text{PlayGolf}, \text{Temperature})$   
 $= 0.94 - 0.911 = \mathbf{0.029}$
- $\text{Gain}(\text{PlayGolf}, \text{Humidity}) = \text{Entropy}(\text{PlayGolf}) - \text{Entropy}(\text{PlayGolf}, \text{Humidity})$   
 $= 0.94 - 0.788 = \mathbf{0.152}$
- $\text{Gain}(\text{PlayGolf}, \text{Windy}) = \text{Entropy}(\text{PlayGolf}) - \text{Entropy}(\text{PlayGolf}, \text{Windy})$   
 $= 0.94 - 0.892 = \mathbf{0.048}$

# Example

| Outlook  | Temperature | Humidity | Windy | Play Golf |
|----------|-------------|----------|-------|-----------|
| Sunny    | Mild        | Normal   | FALSE | Yes       |
| Sunny    | Mild        | High     | FALSE | Yes       |
| Sunny    | Cool        | Normal   | FALSE | Yes       |
| Sunny    | Cool        | Normal   | TRUE  | No        |
| Sunny    | Mild        | High     | TRUE  | No        |
| Overcast | Hot         | High     | FALSE | Yes       |
| Overcast | Mild        | High     | TRUE  | Yes       |
| Overcast | Hot         | Normal   | FALSE | Yes       |
| Overcast | Cool        | Normal   | TRUE  | Yes       |
| Rainy    | Hot         | High     | FALSE | No        |
| Rainy    | Hot         | High     | TRUE  | No        |
| Rainy    | Mild        | High     | FALSE | No        |
| Rainy    | Cool        | Normal   | FALSE | Yes       |
| Rainy    | Mild        | Normal   | TRUE  | Yes       |

# Thank you